

traps or hostile units. He won't fire unless absolutely necessary, rather relying on combat robots or UCAVs (Unmanned Combat Aerial Vehicles) to do the job. Once the target has been identified, the soldier will deploy robots or UCAVs to destroy it.

The individual soldier thus becomes a powerful component of a huge grid, much more powerful than the soldier of today, but also with his safety ensured.

Network-Centric Warfare

August 17, 2016: A pilot flying a fighter jet on a routine patrol spots a group of intruders trying to sneak in through the border. He immediately updates their coordinates on a digital map on his aircraft computer. That done, the information is updated in real-time to army units in the vicinity.

Moving in to intercept the intruders, they get a constant video feed from the aircraft, so they have all they need to know—how many of them there are, what are they currently doing, which way they're going. Using this information, the unit chooses the best spot to ambush the trouble-makers. Ambush and capture them they do, without a single casualty.

The scenario above is a classic one, where the ability to conduct Network-Centric Warfare (NCW) not only increases the efficiency of a unit, it also minimises the loss of men and materiel (equipment).

The term "Network-Centric Warfare" is the latest buzzword in military circles; it is the transition of military forces from the industrial age—where better equipment was the key to success—to the information age, where having the latest information is the key. The idea is to provide real-time, accurate information available to all friendly forces so as to make fast, effective, better-informed decisions.

The All-Important Network

We live in a networked world, be it the cell phone network or the Internet. It was only a matter of time before the network entered the battlefield.

Visualise a huge network of computers with three key components:

Sensors: These include objects that generate useful information (such as pictures or radar images). The objects can include satellites that provide images of a particular area, Unmanned Aerial Vehicles (UAVs) that generate real-time images of an area, and so forth.

Commanders: Military commanders get a broad view of what's going on by collating the information sent by the sensors. They make decisions depending on the situation.

Shooters: Troops on the battlefield that do the actual shooting. It could be a squad of soldiers, robots, tanks, aircraft, or even a single ship. These are guided by the commanders to perform set actions.

These components are analogous to hubs on the network, and the computers are like the nodes. All the computers can access the information flowing around the network. This network relies on a high-bandwidth link using optical fibre or satellite links.

Such networks have to be much more secure than your regular SSL-encrypted connections, due to the highly sensitive nature of the information being transmitted. As a standard procedure, the networks have multiple redundant links, so in the eventuality of war, even if one link is damaged, the network can rely on the others without affecting operations.

In Conventional Warfare

In a more conventional setup, communication happens via radio. Radio communication can not only be intercepted by the enemy, it is also

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